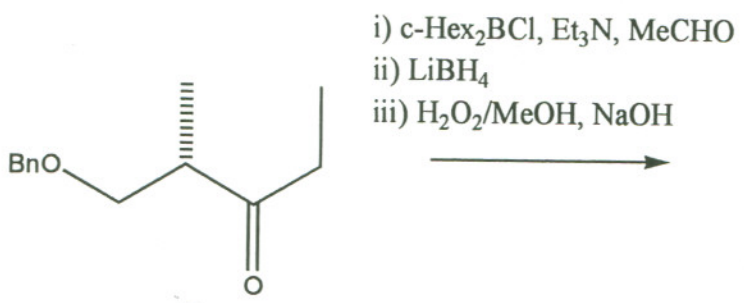
Organic Cumulative Exam October ~~4~~ 5, 2001

The following problems are all from this semester's organic seminar. Choose any 4 of the 6 problems and draw the mechanism. Indicate stereochemistry where appropriate. If you choose reactions 1 and/or 2, you must draw the transition state of the reaction. Be sure to draw clear, unambiguous structures with correct bond angles.

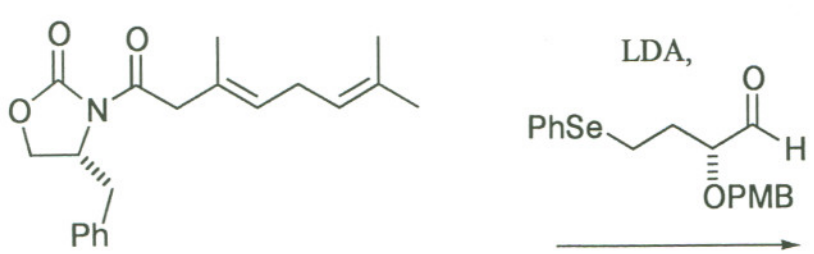
1



Product (?)

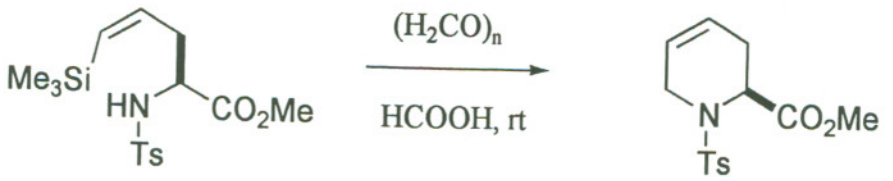
86%, >97% ds

2

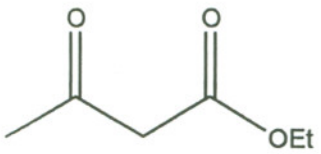


product (?)

(3)



(4)



1) 2.3 LDA
-78°C to 20°C, 14 h

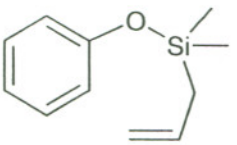
2)

68°C, 14 h

Product (?)

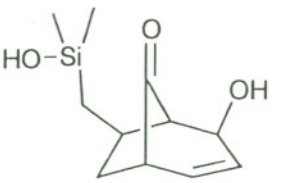
75%

(5)

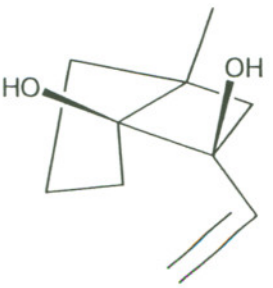


(i) hv

(ii) m-CPBA



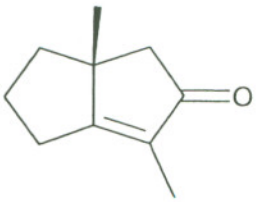
(6)



2 eq. KH

2 eq. 18-C-6

THF, rt



Physical Cumulative Exam
October 5, 2001

A proposed mechanism for the reaction between H_2(gas) and Br_2(gas) is :

$$Br_2 \underset{k_5}{\overset{k_1}{\rightleftharpoons}} 2Br$$

$$Br + H_2 \xrightarrow{k_2} HBr + H$$

$$H + Br_2 \xrightarrow{k_3} HBr + Br$$

$$H + HBr \xrightarrow{k_4} H_2 + Br$$

Determined the rate expression for the rate of production of HBr, [d(HBr)/dt], in terms of rate constants and the concentration of H_2 and Br_2. Under what conditions could one determine the order of the reaction with respect to H_2 and Br_2? What, then, would be the order with respect to H_2 and Br_2?